25900258 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBOT201 - HERBAL TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer

Answer any **10** questions

Each question carries **2** marks

1. Name any four plants maintained in the homegardens of Kerala [K] / [CO1]
2. Mention the advantages of microwave assisted extraction of phytochemicals. [K] / [CO4]
3. Name a commonly used herbal plant and mention its therapeutic application as a home remedy. [K] / [CO2]
4. How is MBG involved in research on medicinal and aromatic plants? [U] / [CO1, CO2]
5. What is phytochemistry? [U] / [CO1]
6. Mention any two advantages of herbal dyes. [U] / [CO4]
7. What is swarming [K] / [CO4]
8. What is the fundamental principle on which solvent extraction is based? [U] / [CO4]
9. Name an equipment used in extraction procedure? [U] / [CO1, CO2]

10. Name any two herbal gardens maintained by Research Centres or Educational institutions [K] / [CO1]

11. Judge the impact of use of synthetic chemicals in herbal plant cultivation. [U] / [CO2]

12. Define transesterification. [U] / [CO4]

[2x10 = 20]

Part B

Short Essay

Answer any **8** questions

Each question carries **5** marks

13. Discuss the sources of herbs. [U] / [CO1]

14. Comment on the characters of a flower to attract honey bees for pollination [U] / [CO4]

15. Explain how adding soap to the tobacco decoction improves its effectiveness as a pesticide. [U] / [CO4]

16. Explain the occurrence, properties, and uses of secondary metabolites in medicine. [K] / [CO3, CO5]

17. Explain applications of herbal dyes. [K] / [CO4]

18. Explain the role of home gardens and botanical gardens in the conservation of herbal plants. [U] / [CO2]

19. Explain the properties of supercritical fluids used in the extraction of secondary metabolites. [U] / [CO4]

20. Clarify the role of soap solution in neem decoction formulation. [U] / [CO4]

21. How does herbal technology promote eco-friendly and sustainable healthcare practices? [K] / [CO1]

22. Analyze the multifaceted role of the Malabar Botanical Garden as a centre for conservation, research and education. [An] / [CO2]

[5x8 = 40]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

23. Discuss the advantages and limitations of using alcohol and acetone in plant extraction methods, with suitable examples. [K] / [CO4]

24. Describe biopesticides and explain their significance. [U] / [CO4]

[10x1 = 10]